

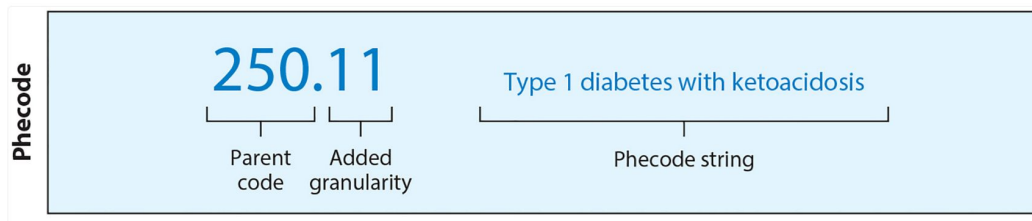
ICD-Map: A Reproducible ICD-Code-to-Phenotype Pipeline

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EHRs Are a Great Source of Phenotypic Data

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- Phecodes: manually curated groupings of ICD codes to efficiently categorize details in Electronic Health Records
- Phecode system was built on ICD-9; ICD-10 & ICD-10-CM codes have been “re-mapped” to phecodes

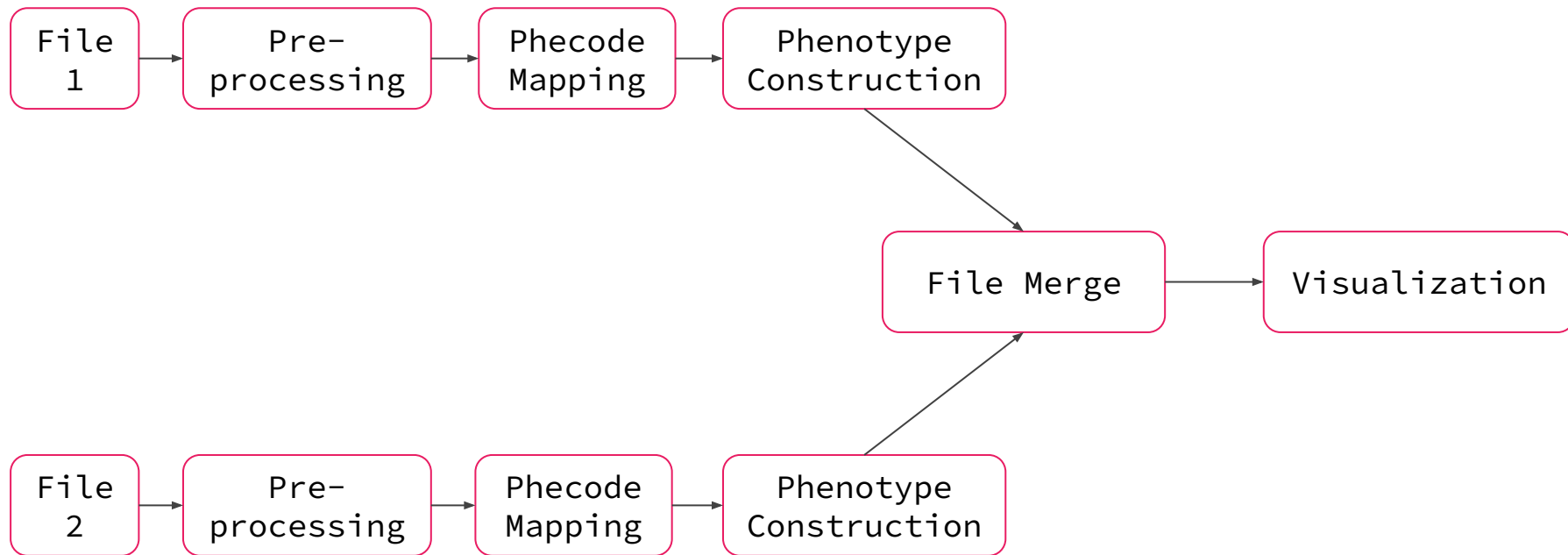


ICD Codes

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- International Classification of Diseases: globally-used medical billing codes regulated by the WHO
 - ICD-X-**CM**: Clinical Modification
 - U.S. version of ICD codes
 - ICD-9: published in 1977; globally adopted in the 1980s
 - ~17,000 codes
 - ICD-9-CM: adopted in the the late 1970s
 - ICD-10: published in 1990; field-tested in the U.S. in 2003
 - >155,000 codes
 - ICD-10-CM: adopted in 2015
 - ICD-10-PCS: codes specific to inpatient settings

Pipeline Architecture



Tools & Methods

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- Nextflow: greater familiarity, better for containerization
- Podman: Daemonless & rootless

Structure

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- Main script : Nextflow script that ties all other components together
- Nextflow modules: Nextflow scripts for each step of the pipeline
- Bin scripts: Python scripts incorporated into Nextflow modules

Reproducibility

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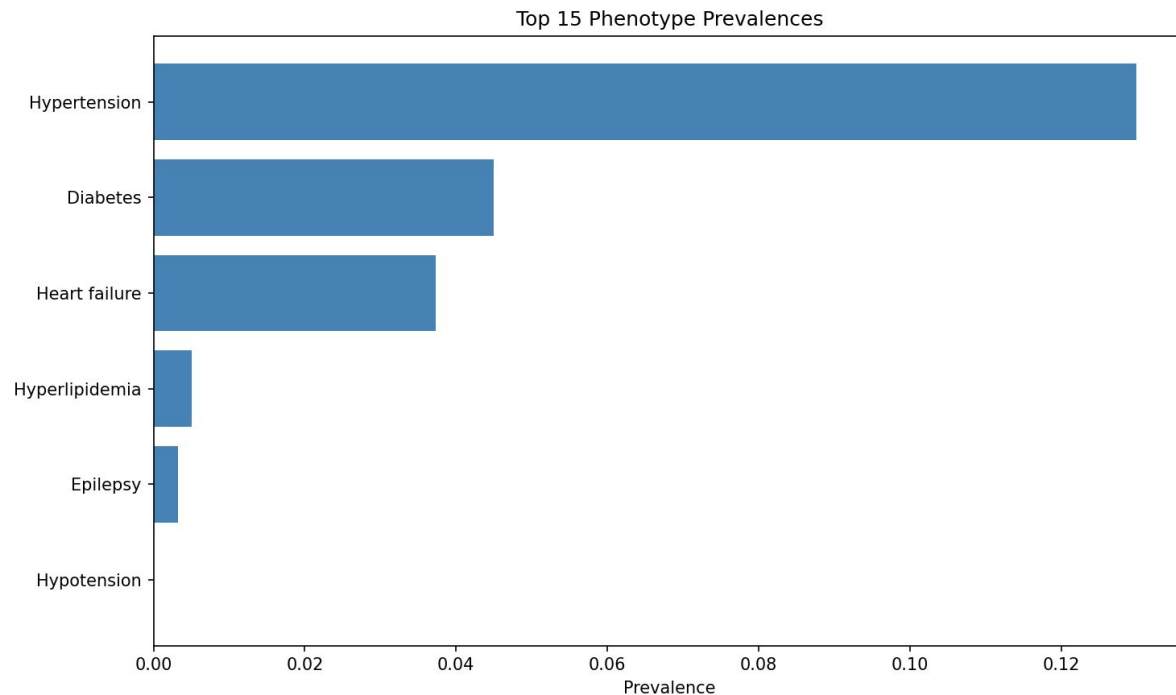
- Podman Containerfile
- Config file for main Nextflow script

Live Demo

- GitHub repository: <https://github.com/zmehta-gt/ICD-Map>

Result & Interpretation

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Limitations & Future Work

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- Adjust workflow for ICD-11 once released